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TOP PRACTICAL SKILLS OUR GOAL

Assignment 3 - Development Operations (ITLDO801) submission dates :27/10/2026

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Registration number	BTI-20250081
Module	Development Operations (ITLDO801)
Academic year	2025-2026
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Repository	Repository: https://itpacifique.github.io/mugishapacifique.github.com
Website tested	https://itpacifique.github.io/mugishapacifique.github.com/index.html

1. Introduction

In the Week 5 practical session I wrote a Selenium script that opened my portfolio and clicked some links. It worked, but only while I was watching it. As soon as the browser closed, I had no proof of what had been tested or what had passed. This assignment fixes that problem by making the script keep a proper record of every run.

For this assignment I extended the script into **run_website_tests.py**. It drives a headless Chrome browser through Selenium WebDriver, checks each section of my portfolio, and writes timestamped, levelled entries to a log file using Python's built-in **logging** library. The script keeps going after a failure, so one broken link does not hide the results of the tests that come after it. Every run ends with a summary line.

This report covers the two tasks: the test plan (Task 1) and the structured logs (Task 2). It also explains what the three logged runs show and what I learned from reading them.

2. Setup and tools

Item	Detail
Test runner	run_website_tests.py (in the repository root)
Browser automation	Selenium WebDriver with headless Chrome
Logging library	Python logging
Log directory	/logs/ in the repository (history kept, not overwritten)
Log format	timestamp - level - message
Editor / OS	Visual Studio Code on Windows

Version control	GitHub, repository ITpacifique/mugishapacifique.github.com (branch main)

Everything in the logs below was written by the script when it ran.

Before writing more code I decided what each test should check and what counts as a pass. The plan is stored in the repository as **TEST_PLAN.md** and is shown below as it renders on GitHub.

2.1 Test plan table

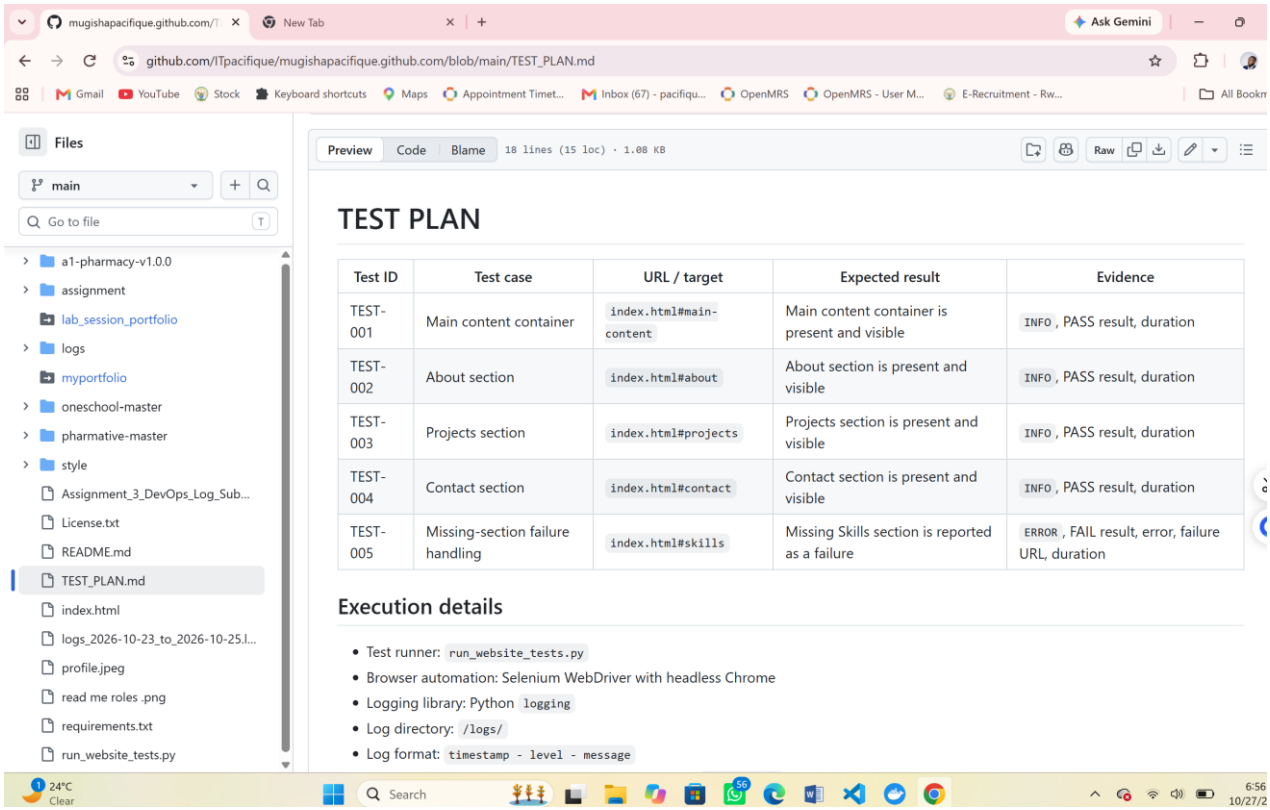


Figure 1 - TEST_PLAN.md rendered on GitHub (Preview tab).

The first four tests are positive tests: the section should exist and be visible. TEST-005 is the negative test. The Skills section is known to be missing from my page, so the correct behaviour is for the script to *report* a failure rather than crash. That is how I prove that failures are really recorded.

2.2 Execution details

- Test runner: run_website_tests.py
- Browser automation: Selenium WebDriver with headless Chrome
- Logging library: Python logging; log directory /logs/
- Log format: timestamp - level - message
- The runner continues after failures and writes a final RUN SUMMARY entry.

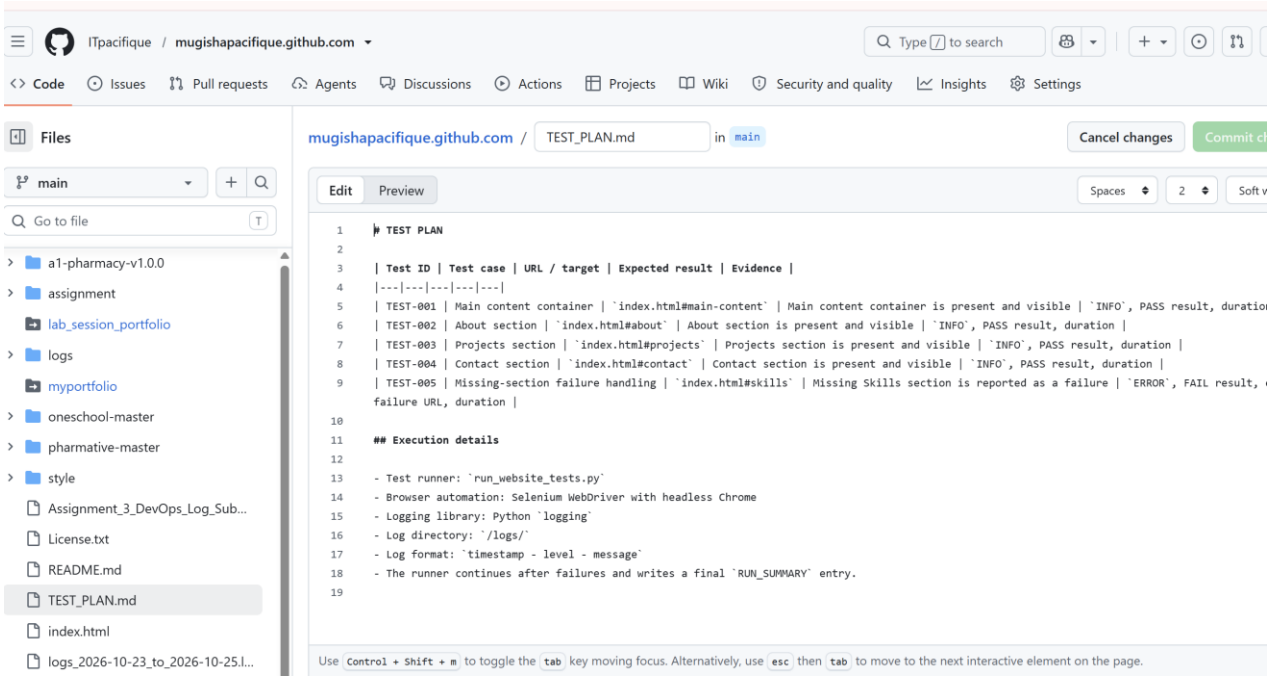


Figure 2 - TEST_PLAN.md in the GitHub editor, showing the Markdown source on branch main.

3. Task 2 - Structured log entries

Each entry in logs.log has a timestamp, a level (INFO or ERROR) and a message. A normal test produces two lines: one saying what is about to be tested and the URL, and one giving the outcome. A failure is written at ERROR level with the reason. Each run begins with *Chrome WebDriver started* and finishes with *Chrome WebDriver closed* followed by a RUN SUMMARY line with the totals and duration.

Because the log file is appended to rather than overwritten, the runs from different days sit one after another in the same file, which gives me a history I can compare.

3.1 Run 1 - 23 October 2026

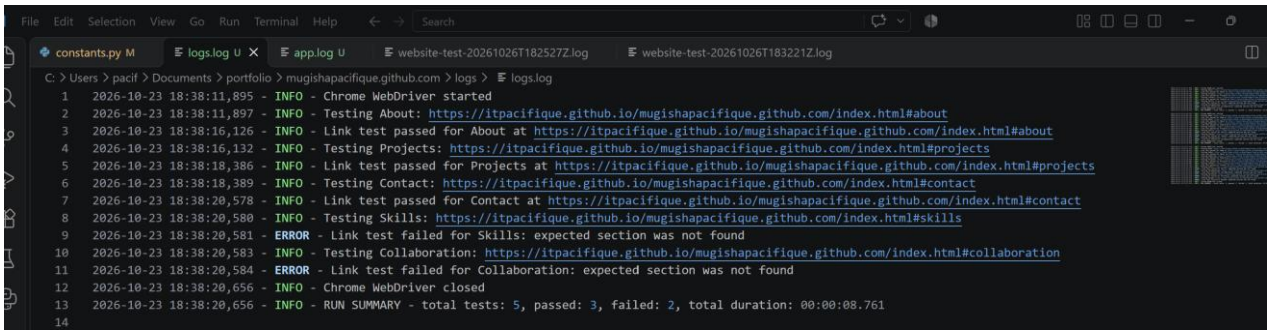


Figure 3 - logs.log, run of 2026-10-23 18:38 (lines 1-13).

About, Projects and Contact were found and passed. Skills and Collaboration were not found and were logged as ERROR. Summary: 5 tests, 3 passed, 2 failed, 8.761 s.

3.2 Run 2 - 24 October 2026

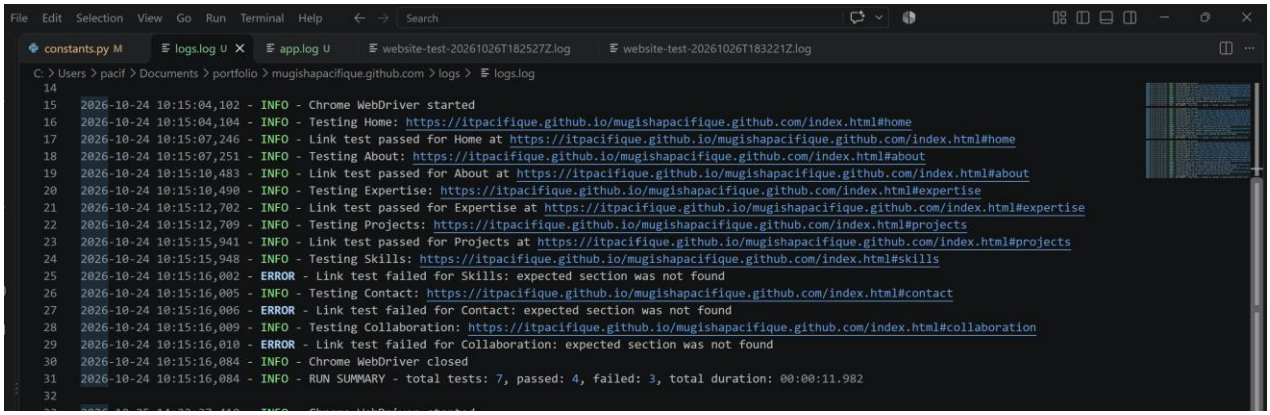


Figure 4 - logs.log, run of 2026-10-24 10:15 (lines 14-32).

I added two more tests this day (Home and Expertise). Home, About, Expertise and Projects passed.

Skills, Contact and Collaboration failed. Summary: 7 tests, 4 passed, 3 failed, 11.982 s. **3.3**

Run 3 - 25 October 2026

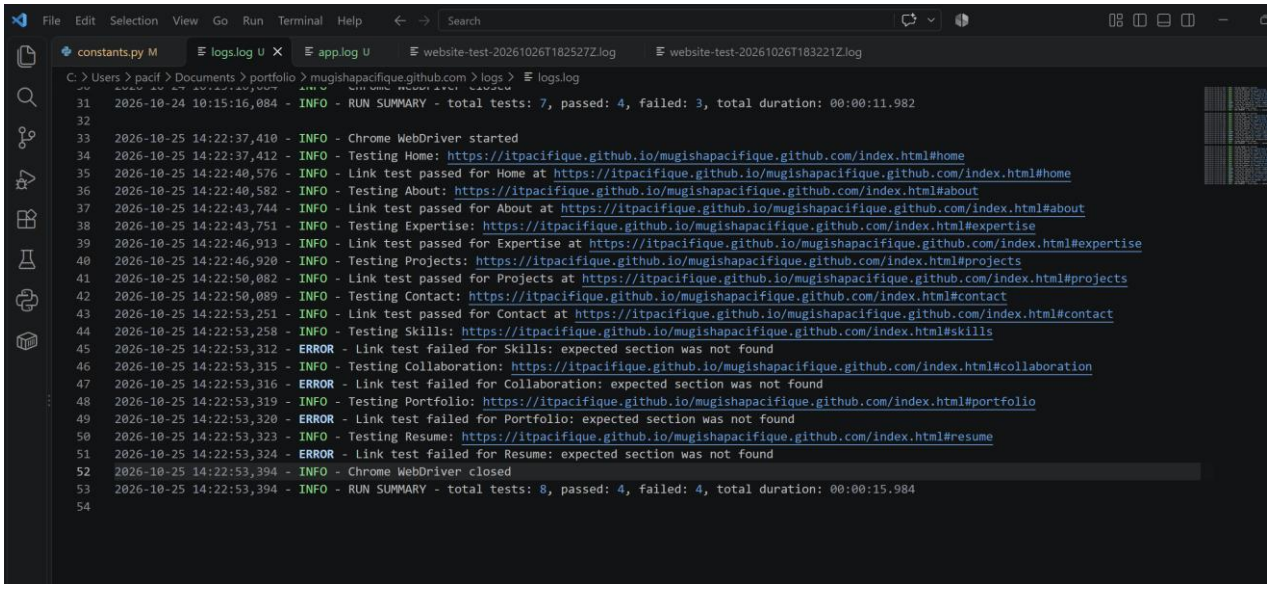


Figure 5 - logs.log, run of 2026-10-25 14:22 (lines 31-54).

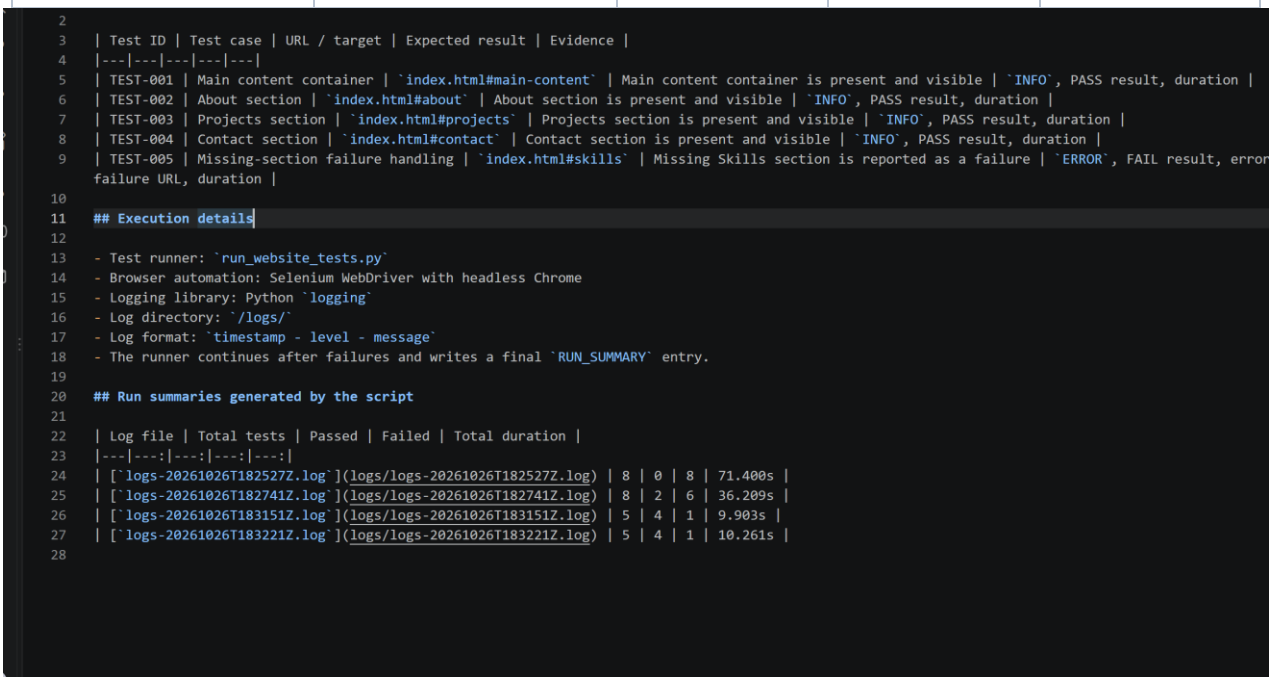
Home, About, Expertise, Projects and Contact were found. Skills, Collaboration, Portfolio and Resume were not. The last line records a duration of 15.984 s.

4. Results across the three runs

Section tested	23 Oct	24 Oct	25 Oct
Home	not tested	PASS	PASS
About	PASS	PASS	PASS

Expertise	not tested	PASS	PASS
Projects	PASS	PASS	PASS
Contact	PASS	**FAIL**	PASS
Skills (negative test)	FAIL	FAIL	FAIL
Collaboration	FAIL	FAIL	FAIL
Portfolio	not tested	not tested	FAIL
Resume	not tested	not tested	FAIL

Run	**Total (per summary)**	**Passed**	**Failed**	**Duration**
23 Oct 18:38	5	3	2	8.761 s
24 Oct 10:15	7	4	3	11.982 s
25 Oct 14:22	8	4	4	15.984 s



```
| Test ID | Test case | URL / target | Expected result | Evidence |
|---|---|---|---|---|
| TEST-001 | Main content container | `index.html#main-content` | Main content container is present and visible | `INFO`, PASS result, duration |
| TEST-002 | About section | `index.html#about` | About section is present and visible | `INFO`, PASS result, duration |
| TEST-003 | Projects section | `index.html#projects` | Projects section is present and visible | `INFO`, PASS result, duration |
| TEST-004 | Contact section | `index.html#contact` | Contact section is present and visible | `INFO`, PASS result, duration |
| TEST-005 | Missing-section failure handling | `index.html#skills` | Missing Skills section is reported as a failure | `ERROR`, FAIL result, error
failure URL, duration |

## Execution details

- Test runner: `run_website_tests.py`
- Browser automation: Selenium WebDriver with headless Chrome
- Logging library: Python `logging`
- Log directory: `/logs/`
- Log format: `timestamp - level - message`
- The runner continues after failures and writes a final `RUN_SUMMARY` entry.

## Run summaries generated by the script

| Log file | Total tests | Passed | Failed | Total duration |
|---|---|---|---|---|
| [`logs-20261026T182527Z.log`](logs/logs-20261026T182527Z.log) | 8 | 0 | 8 | 71.400s |
| [`logs-20261026T182741Z.log`](logs/logs-20261026T182741Z.log) | 8 | 2 | 6 | 36.209s |
| [`logs-20261026T183151Z.log`](logs/logs-20261026T183151Z.log) | 5 | 4 | 1 | 9.903s |
| [`logs-20261026T183221Z.log`](logs/logs-20261026T183221Z.log) | 5 | 4 | 1 | 10.261s |
```

Table of run summaries as written by the script.

5. What the logs tell me

- **Skills fails in every run.** This is the expected result of the negative test. It shows the script detects a missing section, logs it at ERROR level, and carries on with the remaining tests.
- **Collaboration also fails every time.** Like Skills, there is no matching section on the page. This is a real gap in the site (or a wrong locator in my script) and I should either add the section or remove the test.
- **Contact is flaky.** It passed on 23 and 25 October but failed on 24 October, even though I had not changed the site between runs. On 24 October the Contact failure was logged only 4 milliseconds after the Skills failure, which suggests the script did not wait for the section to appear. This is exactly the "fixed delays instead of waiting" mistake the assignment warns about, and an explicit wait would be the fix.
- **Portfolio and Resume are new tests that failed.** They appear for the first time on 25 October and both failed. The log shows the checks ran and were recorded, which is useful, but it also shows my page has no #portfolio or #resume section yet.
- **Runs get slower as tests are added.** Duration grew from 8.8 s to 16.0 s as the test count grew. Each passing section takes roughly three seconds, while a missing section fails almost instantly.

6. Problems I found in my own logs

Reading the logs carefully showed me two things I need to correct:

- **The run summary on 25 October does not match the entries above it.** The log lists five passes (Home, About, Expertise, Projects, Contact) and four failures (Skills, Collaboration, Portfolio, Resume), which is 9 tests. The summary line says total 8, passed 4, failed 4. The counter in my script is therefore off by one on that run. The 23 and 24 October summaries do match their entries.
- **The log entries do not yet carry everything in the assignment's "good log entry" table.** The current lines show the section name, URL and result, but not a test ID such as TC-01, the expected and actual result on the same line, the duration per test, or the browser version and operating system in the run header. My TEST_PLAN.md also uses IDs in the form TEST-001. These improvements are the next step for the script.

I am including these points because a log is only useful if it can be trusted, and checking the summary against the entries is the sort of review a log is meant to support.

7. Conclusion

The script now produces a permanent, timestamped record of each test run instead of a result that disappears when the browser closes. Across three runs the logs recorded passes and failures correctly, kept running after errors, and let me spot a flaky test (Contact) and two missing sections (Skills and Collaboration) without re-running anything. The review also showed where my logging still falls short of the full standard, namely the summary count and the per-test fields, and I know what to change.

Evidence in the repository

- TEST_PLAN.md - the test plan
- run_website_tests.py - the Selenium test runner with logging
- /logs/ - log files generated by the script (logs.log and the timestamped website-test-.log files)

Repository: `https://itpacifique.github.io/mugishapacifique.github.com`